Experiment 03 - Build a parity generator and detector.

Learning Objectives: To Minimize the Boolean algebra and design it using logic gates

Instruments:

Sr.No.	Instrument / Component	Range / Value
1	I.Cs.	7408, 7432, 7404, 7400, 7402, 7486.
2	Bread Board	---
3	Connecting Wires	---
4	DMM	---

Theory:

Parity Bit:

The parity generating technique is one of the most widely used error detection techniques for the data transmission. In digital systems, when binary data is transmitted and processed, data may be subjected to noise so that such noise can alter 0s (of data bits) to 1s and 1s to 0s. Hence, a Parity Bit is added to the word containing data in order to make number of 1s either even or odd. The message containing the data bits along with parity bit is transmitted from transmitter to the receiver. At the receiving end, the number of 1s in the message is counted and if it doesn't match with the transmitted one, it means there is an error in the data. Thus, the Parity Bit is used to detect errors, during the transmission of binary data.

Parity Generator:

It is combinational circuit that accepts an n-1 bit stream data and generates the additional bit that is to be transmitted with the bit stream. This additional or extra bit is termed as a parity bit. In even parity bit scheme, the parity bit is '0' if there are even number of 1s in the data stream and the parity bit is '1' if there are odd number of 1s in the data stream. In odd parity bit scheme, the parity bit is '1' if there are even number of 1s in the data stream and the parity bit is '0' if there are odd number of 1s in the data stream. Let us discuss both even and odd parity generators.

Even Parity and Odd Parity:

The sum of the data bits and parity bits can be even or odd. In even parity, the added parity bit will make the total number of 1s an even number, whereas in odd parity, the added parity bit will make the total number of 1s an odd number.

The basic principle involved in the implementation of parity circuits is that the sum of odd numbers of 1s is always 1 and sum of even numbers of 1s is always 0. Such error detecting and correction can be implemented by using Ex-OR gates (since Ex-OR gates produce zero output when there are even numbers of inputs).

Even Parity Generator:

A 3-bit message is to be transmitted with an even parity bit. Let the three inputs A, B and C are applied to the circuit and the output bit is the parity bit P. The total number of 1s must be even, to generate the even parity bit P.

Truth Table:

A	B	C	Even Bit Parity Generator (P)

K-map simplification for 3-bit message even parity generator:

Logic Equation:

Logic Diagram:

Odd Parity Generator:

A 3-bit data is to be transmitted with an odd parity bit. The three inputs are A, B and C and P is the output parity bit. The total number of bits must be odd in order to generate the odd parity bit.

Truth Table:

A	B	C	Odd Bit Parity Generator (P)

K-map simplification for 3-bit message even parity generator:

Logic Equation:

Logic Diagram:

Parity Check

It is a logic circuit that checks for possible errors in the transmission. This circuit can be an even parity checker or odd parity checker depending on the type of parity generated at the transmission end. When this circuit is used as even parity checker, the number of input bits must always be even. When a parity error occurs, the 'sum even' output goes low and 'sum odd' output goes high. If this logic circuit is used as an odd parity checker, the number of input bits should be odd, but if an error occurs the 'sum odd' output goes low and 'sum even' output goes high.

Even Parity Checker

Consider that three input messages along with an even parity bit is generated at the transmitting end. These 4 bits are applied as input to the parity checker circuit which checks the possibility of error on the data. Since the data is transmitted with even parity, four bits received at circuit must have an even number of 1s. If any error occurs, the received message consists of an odd number of 1s. The output of the parity checker is denoted by PEC (parity error check). The below table shows the truth table for the even parity checker in which PEC = 1 if the error occurs, i.e., the four bits received have an odd number of 1s and PEC = 0 if no error occurs, i.e., if the 4-bit message has even number of 1s.

Truth Table:

A	B	C	Parity	PEC

K-map simplification for 3-bit message even parity checker:

Logic Equation:

Logic Diagram:

Odd Parity Checker

Consider that a three bit message along with odd parity bit is transmitted at the transmitting end. Odd parity checker circuit receives these 4 bits and checks whether any error are present in the data. If the total number of 1s in the data is odd, then it indicates no error, whereas if the total number of 1s is even then it indicates the error since the data is transmitted with odd parity at transmitting end. The below figure shows the truth table for odd parity generator where PEC =1 if the 4-bit message received consists of even number of 1s (hence the error occurred) and PEC= 0 if the message contains odd number of 1s (that means no error)

Truth Table:

A	B	C	Parity	PEC

K-map simplification for 3-bit message odd parity checker:

Logic Equation:

Logic Diagram:

Learning Outcomes: The student should have the ability to

LO1: **explain** the functioning of parity generator and checker logic circuits

LO2: **design** parity generator and checker logic circuits

LO3: **test** the output of parity generator and checker logic circuits with the help of truth tables

Course Outcome: After completion of this experiment the students will be able to build a parity generator and detector using logic gates.

Conclusion:

For Faculty Use

Correction Parameters	**Formative Assessment [40%]**	**Timely completion of Practical [40%]**	**Attendance / Learning Attitude [20%]**	
Marks Obtained				